# Tolerance and Autoimmunity

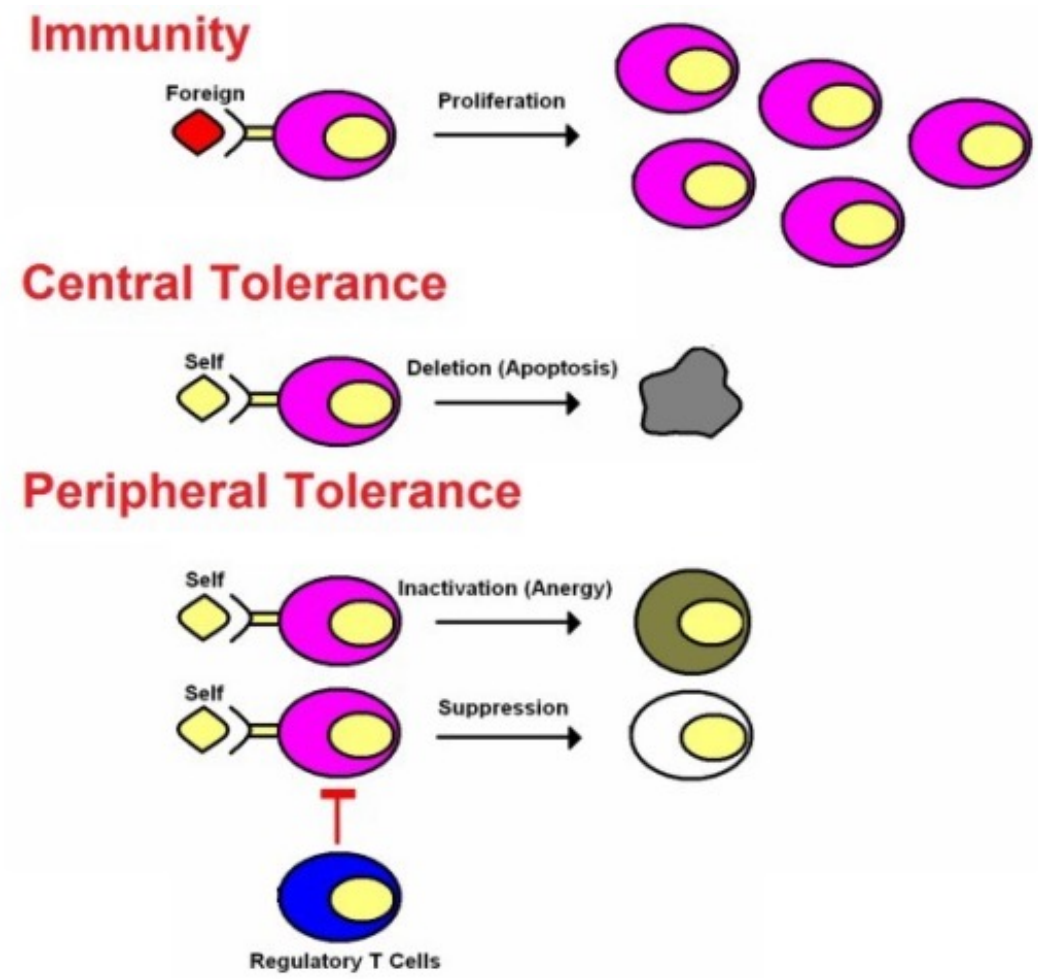
BMS117

Michael Waterfield, MD/PhD

# Immune Tolerance and Autoimmunity

- Diverse antigen-specificity in T and B cells includes reactivity to self-antigens
  - potential danger of self-damage.
- Tolerance : elimination/control of self-reactive T and B cells
  - T cell tolerance & B cell tolerance
- Failure of tolerance
  - Autoimmune diseases (diabetes, Rheumatoid arthritis,…)

# Autoimmune Disease is caused by defective Immune Tolerance



# T cell Central Tolerance

**<u>During T cell development</u>** <u>in the thymus</u>

1. T cells develop from hematopoietic stem cells that reside in the bone marrow and migrate to the thymus - these seed the thymus throughout life.

2. After reaching thymus, these stem cells become T cell precursors (thymocytes) and rearrange their TCR genes, generating diversity.

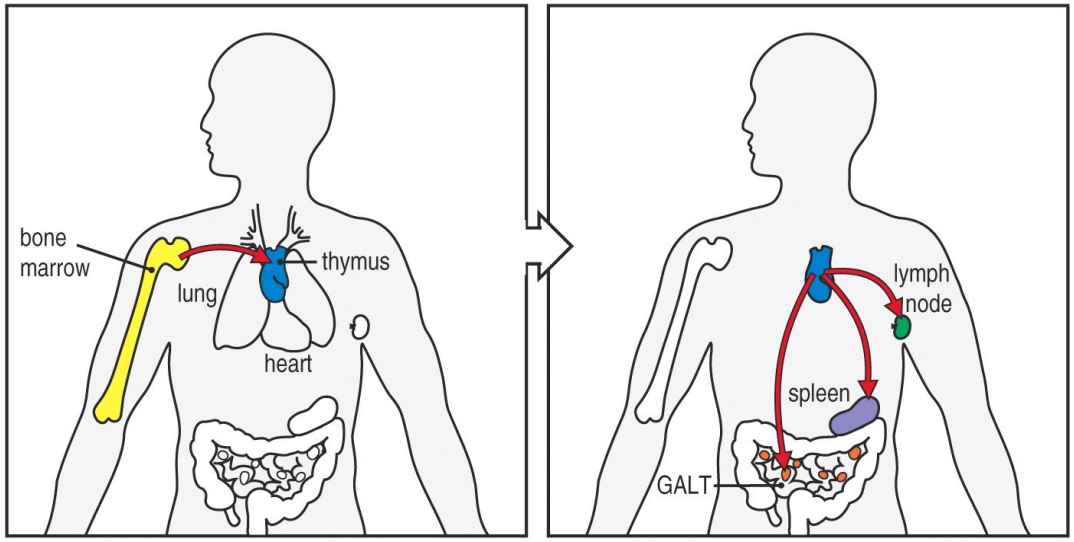
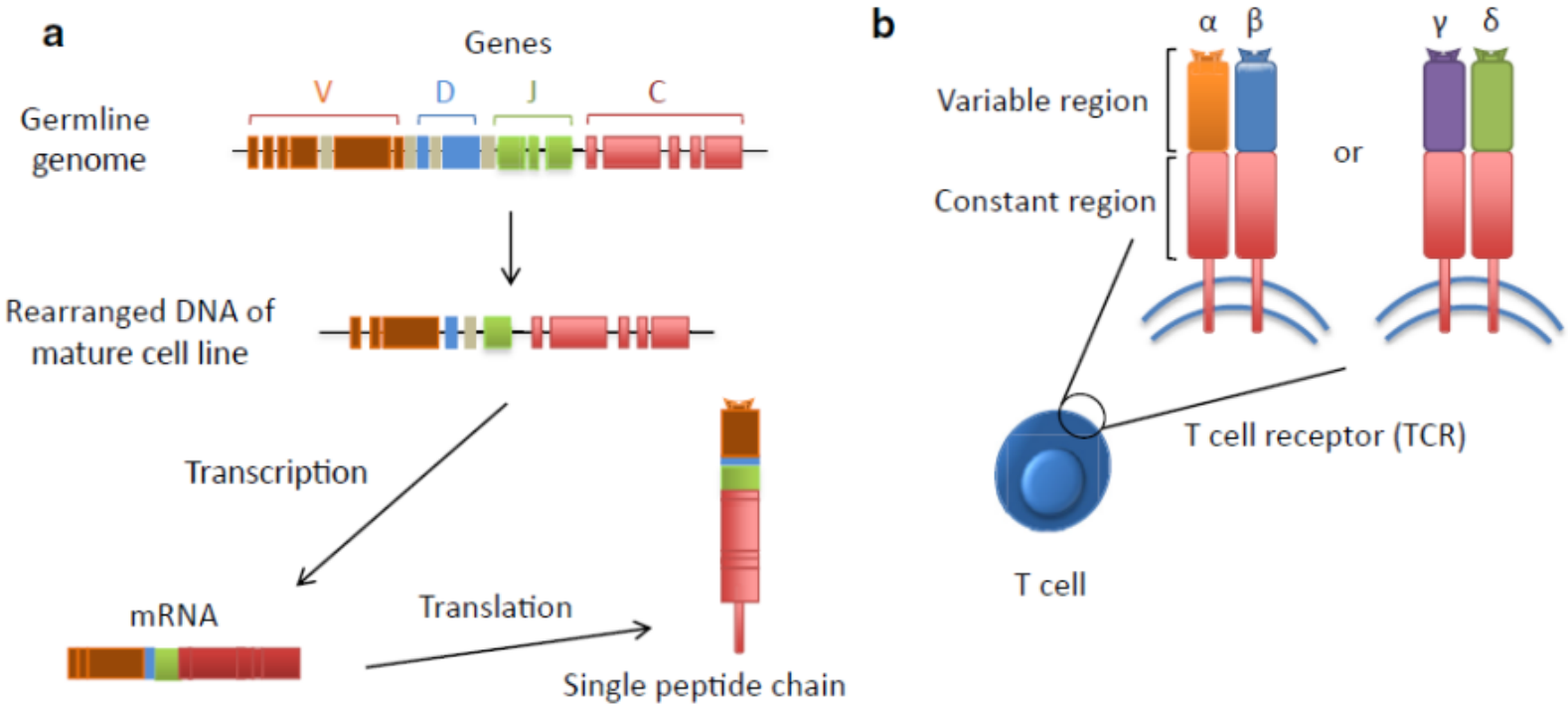



Figure 5-1 The Immune System, 2/e (© Garland Science 2005)

# TCR rearrangement

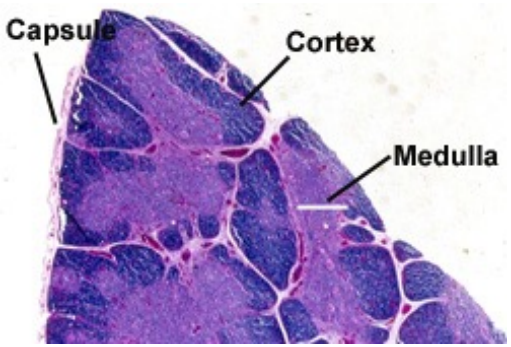



**Due to TCR recombination T cells are capable of tremendous diversity. Some of the T cells generated will react to self-antigens. These need to be purged in the thymus to prevent autoimmunity.**

## Summary of TCR diversity-generating mechanisms

- Each TCR has two chains
- Multiple variable domain exons (V, D, and J)
- Multiple segments of V, D, and J
- Imprecision in the joining process
  - Nucleotides added or excised

# Positive selection




TCR-expressing thymocytes interact with MHC-expressing cortical epithelial cells in the cortex.

- Binding to MHC – survive (**positive selection**)
- No binding to MHC – die (95% of thymocytes)

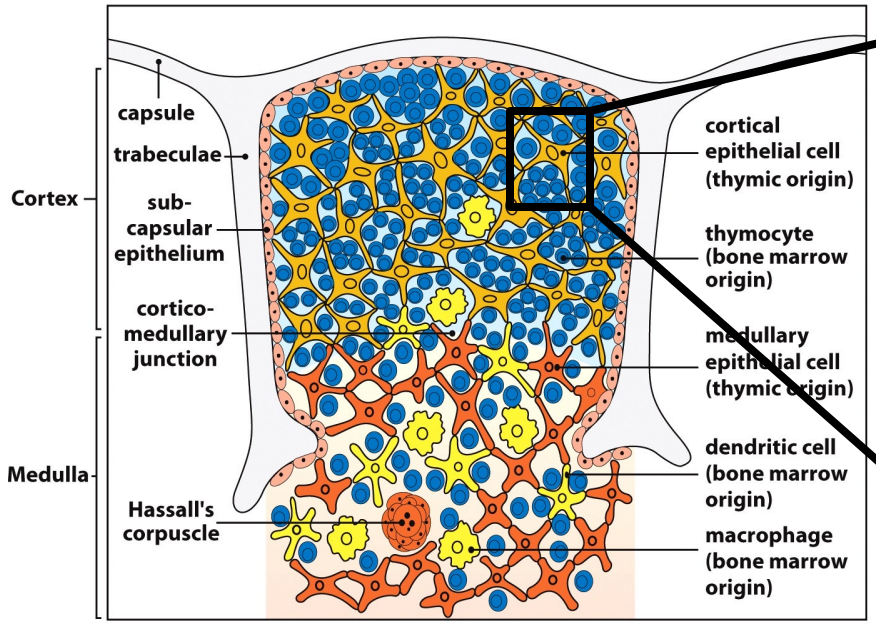
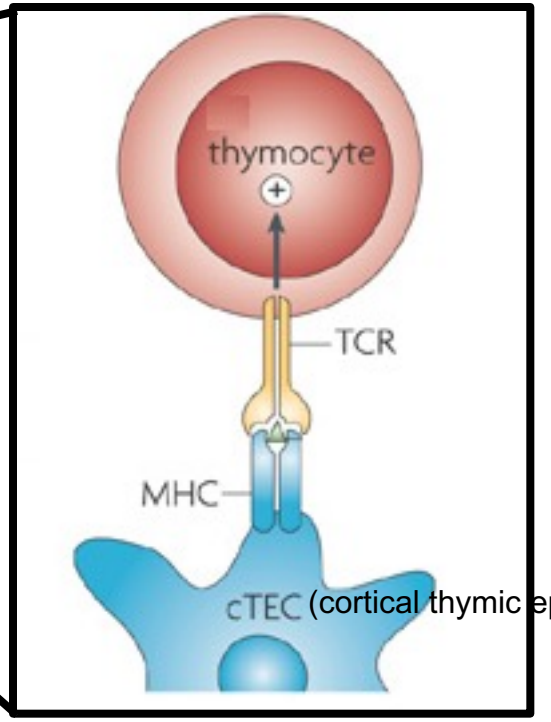



Figure 7.3 part 2 of 2 The Immune System, 3ed. (© Garland Science 2009)



# Negative selection (=Central Tolerance)

Positively selected thymocytes interact with MHC-expressing medullary epithelial cells or dendritic cells in the medulla.

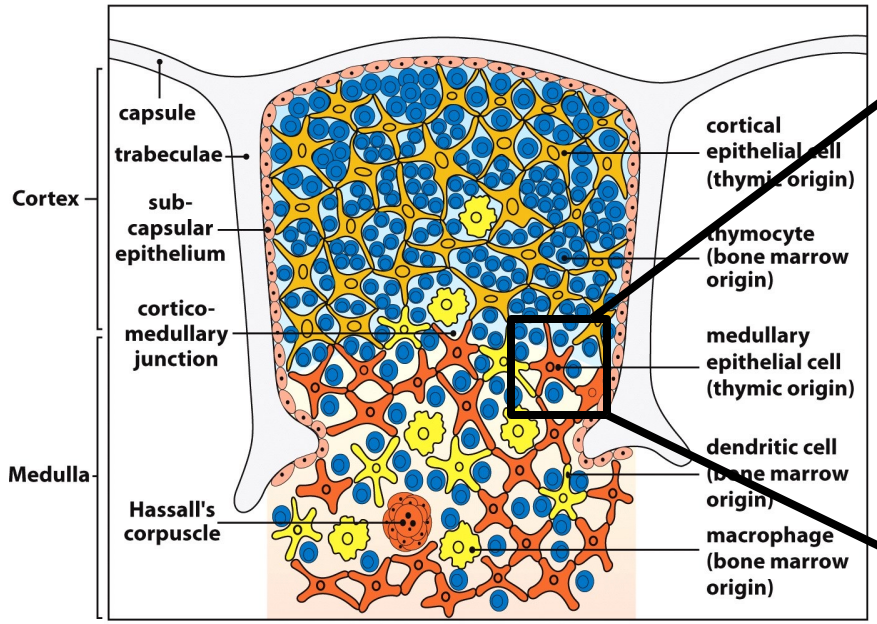
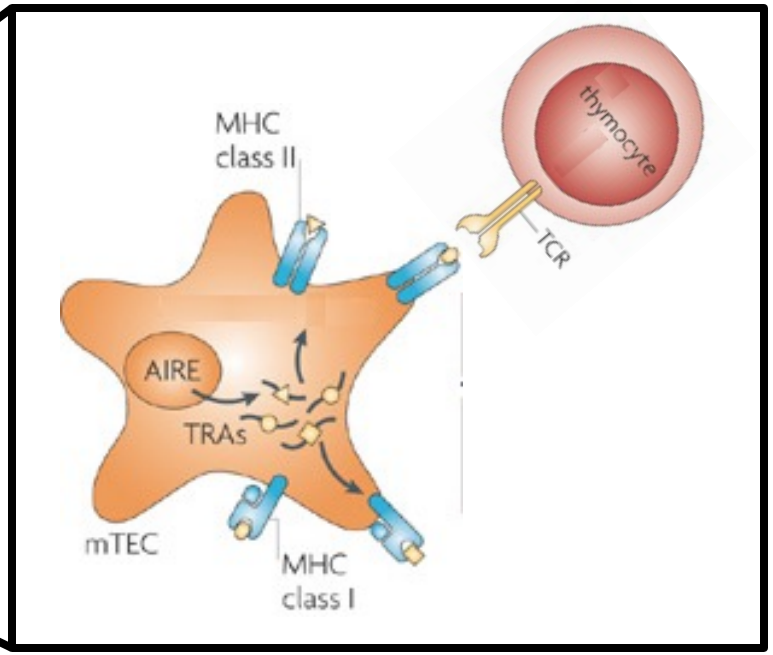
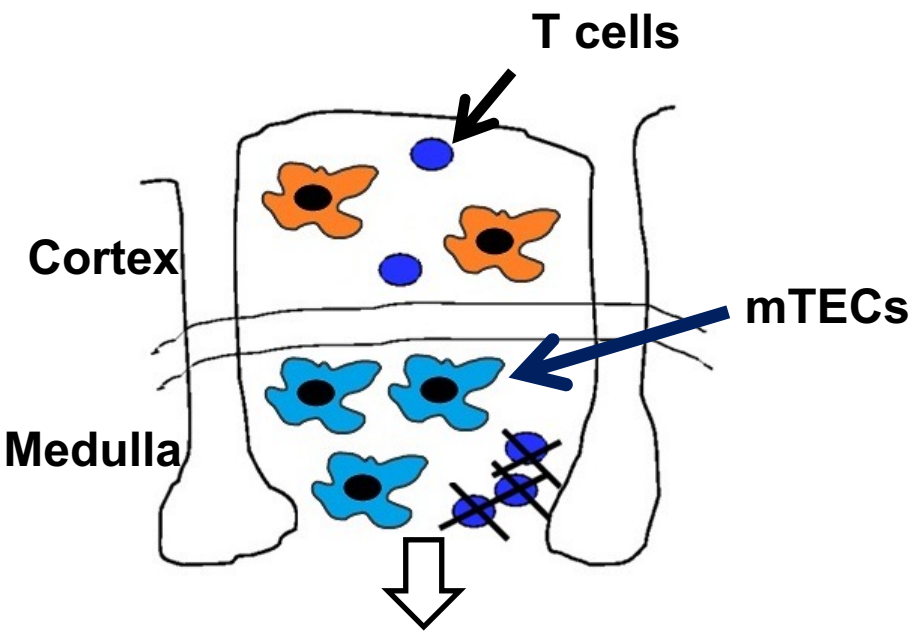



Figure 7.3 part 2 of 2 The Immune System, 3ed. (© Garland Science 2009)



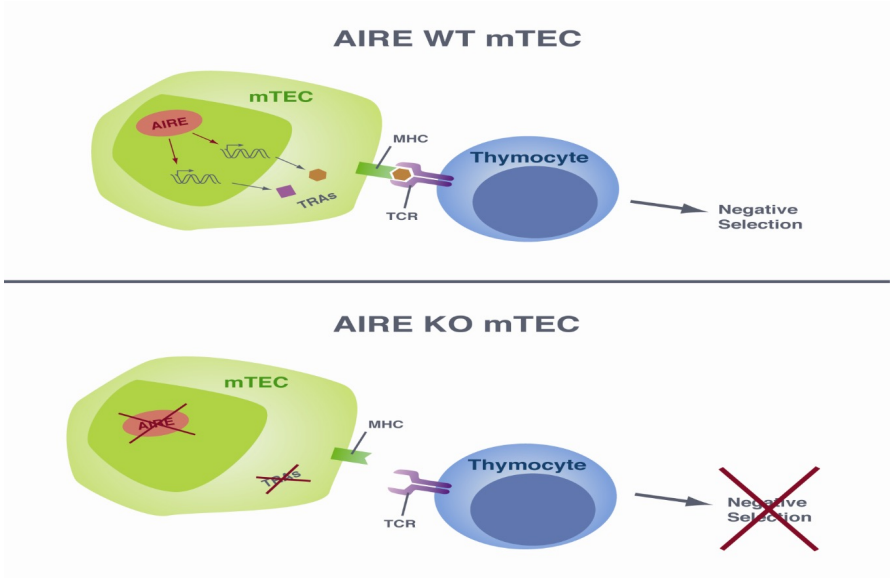

- Medullary epithelial cells express the transcription factor AIRE (AutoImmune REgulator), which leads to the expression and presentation of tissue-restricted self-antigens (TRAs) to thymocytes.

# Medullary thymic epithelial cells (mTECs) mediate deletion of self-reactive thymocytes




Self organ-specific T cells deleted

Positive Selection of Tregs




- Deletion of AIRE leads to APS1/APECED
- mTECs express 80-90% of the protein coding genome
- TRAs (such as insulin) are expressed stochastically

- Dendritic cells are endocytic and thus actively take up mTEC-driven TRAs or blood-born antigens and present them to thymocytes.

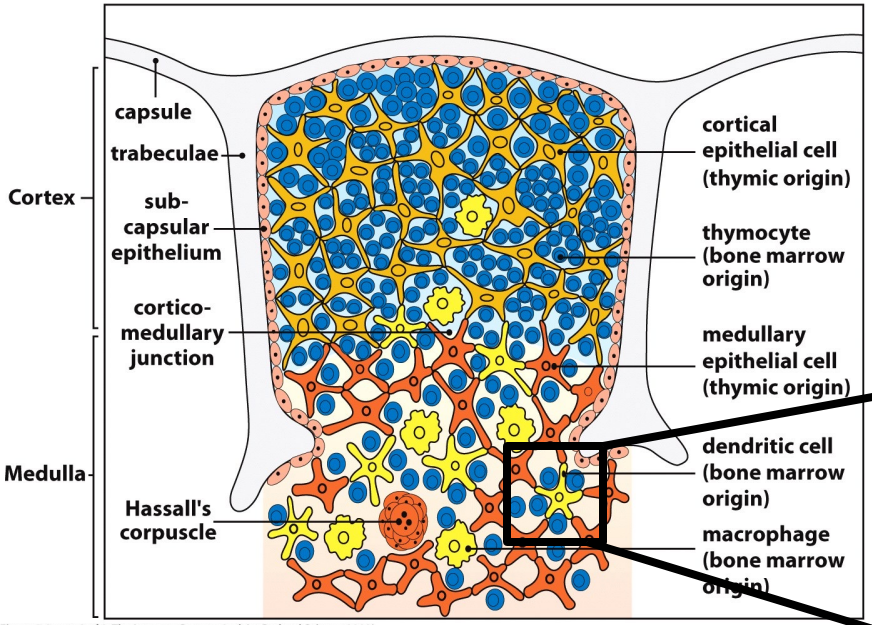
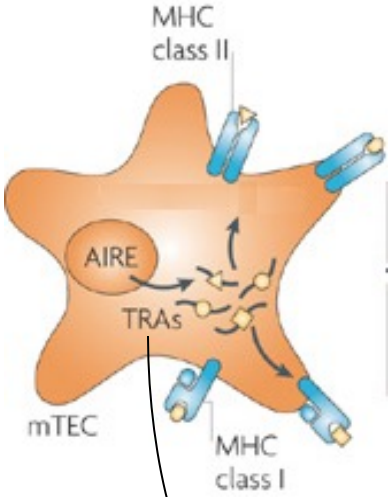
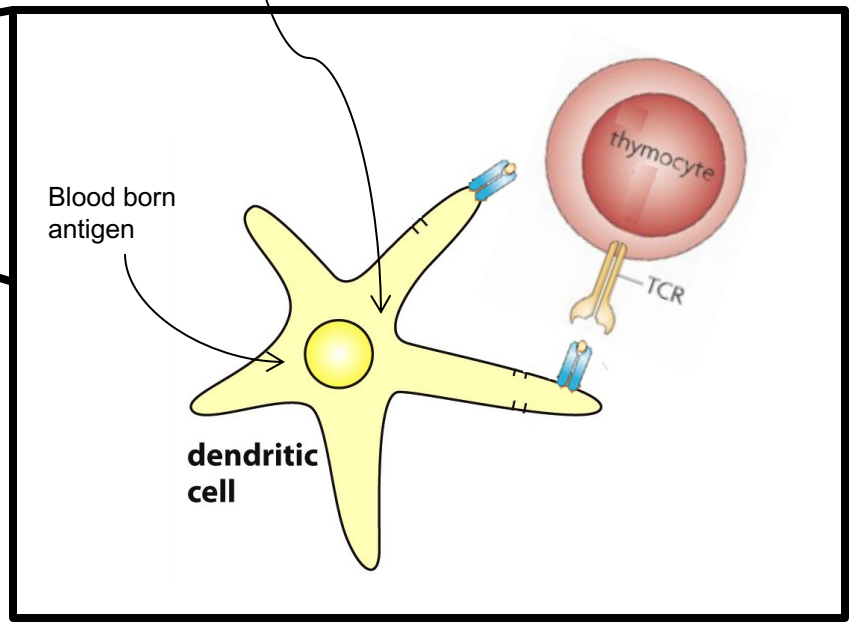


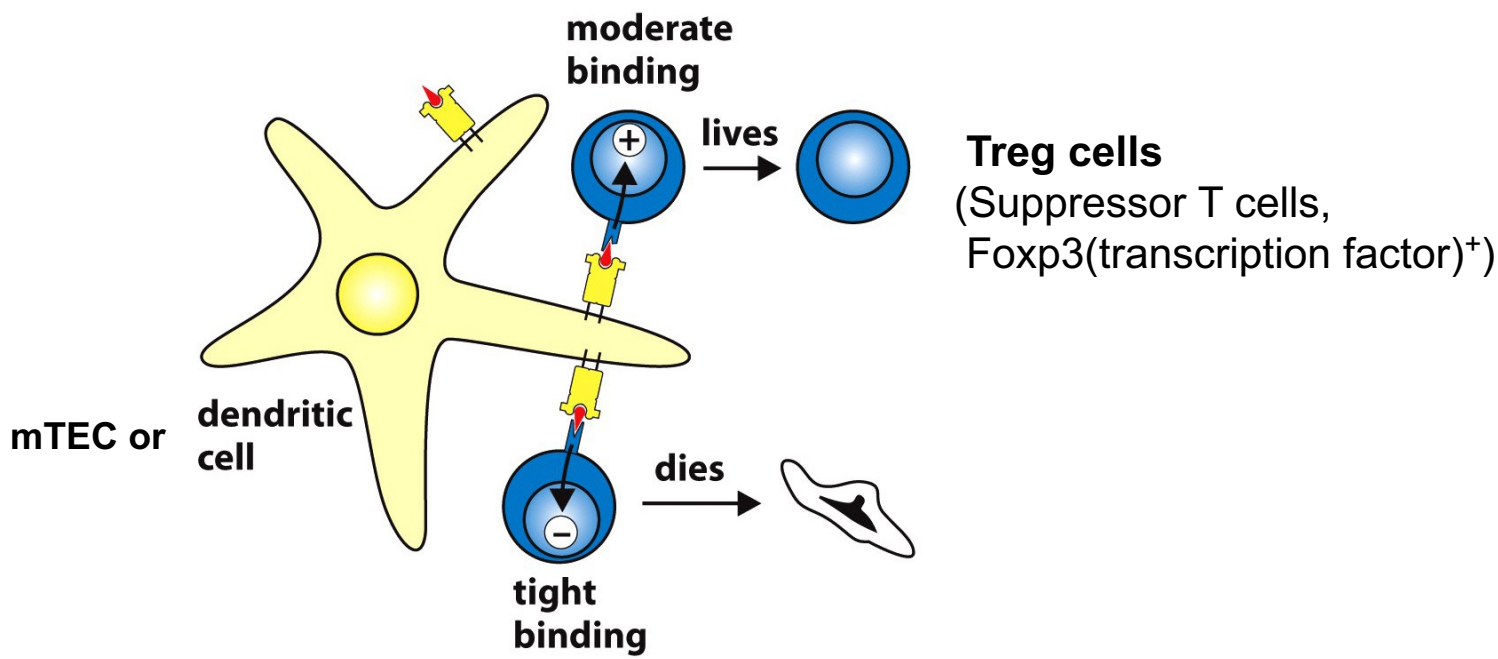

Figure 7.3 part 2 of 2 The Immune System, 3ed. (© Garland Science 2009)



- Thymocytes that interact with mTECs/DCs with strong affinity
  - die (**negative selection**)

- Thymocytes that interact with mTECs/DCs with intermediate affinity
  - differentiate to **regulatory T (Treg) cells**




**AIRE mutation:** Autoimmune Polyendocrinopathy Syndrome type 1 (APS-1)

**Foxp3 mutation**: X-linked IPEX syndrome (Immunodysregulation, Polyendocrinopathy, and Enteropathy)

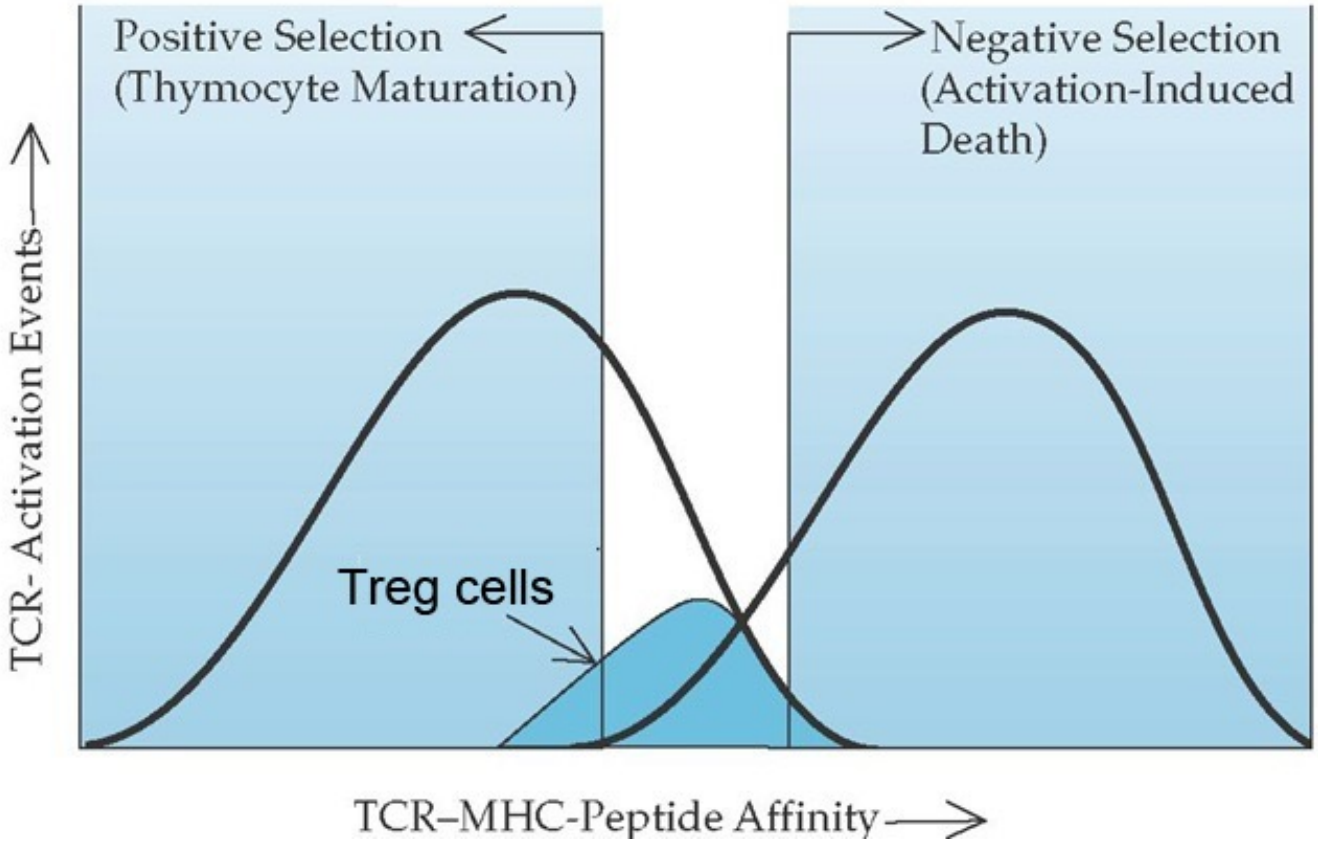
# APECED: an example of failed central (thymic) tolerance

| APECED patients suffer a variety of autoimmune diseases and candidiasis | |
|---|---|
| Symptom | Frequency in Finnish patients (%) |
| **Endocrine glands** | |
| Hypoparathyroidism | 85 |
| Adrenal failure | 72 |
| Ovarian failure | 60 |
| Insulin-dependent diabetes mellitus | 18 |
| Testicular atrophy | 14 |
| Parietal cell atrophy | 13 |
| Hypothyroidism | 6 |
| **Other tissues** | |
| Candidiasis | 100 |
| Dental enamel hypoplasia | 77 |
| Nail dystrophy | 52 |
| Tympanic membrane calcification | 33 |
| Alopecia | 27 |
| Keratopathy | 22 |
| Vitiligo | 13 |
| Hepatitis | 13 |
| Intestinal malabsorption | 10 |

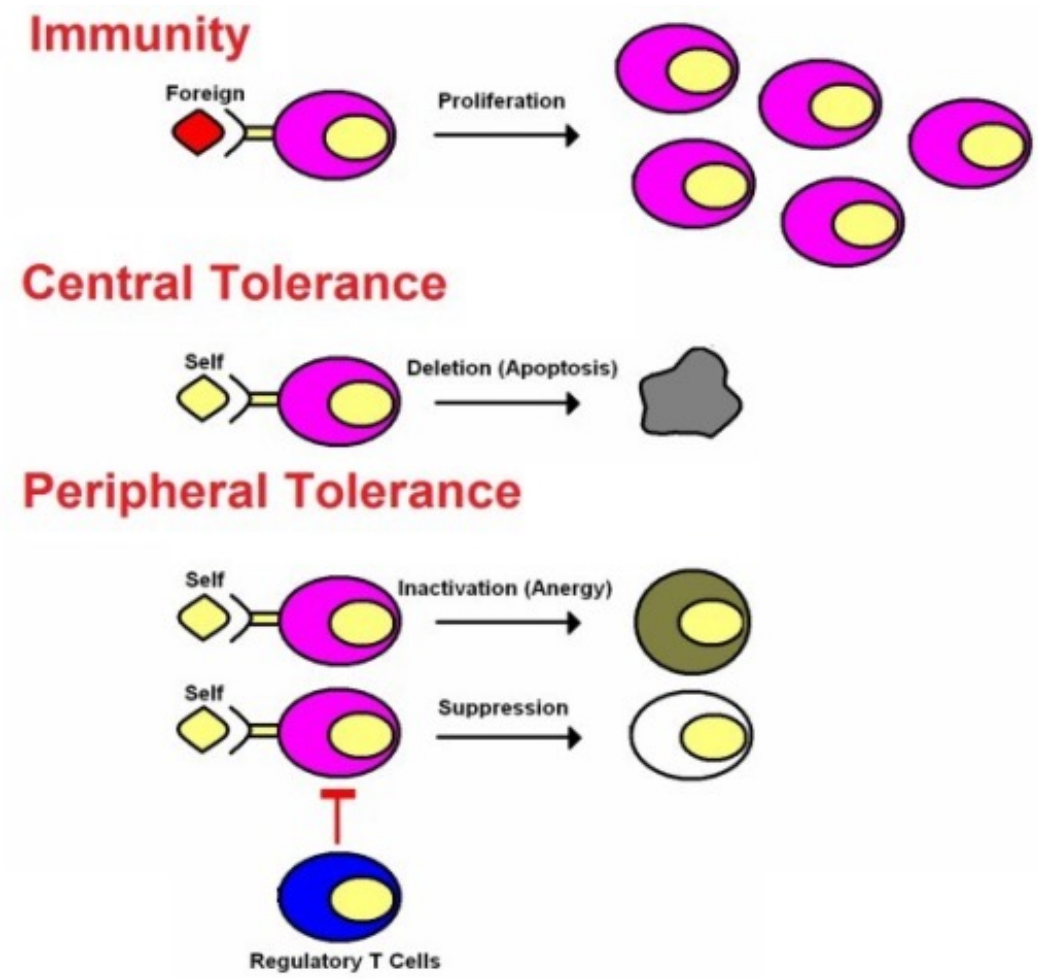
Figure 11-19 The Immune System, 2/e (© Garland Science 2005)

- APECED is caused by a single gene defect
- Defective gene is called Aire (Autoimmune Regulator)
- Aire regulates the expression of self-proteins in the medullary thymic epithelial cells for negative selection of T cells
- SNPs in *AIRE* have been linked to rheumatoid arthritis

# Summary of T cell selection in the thymus

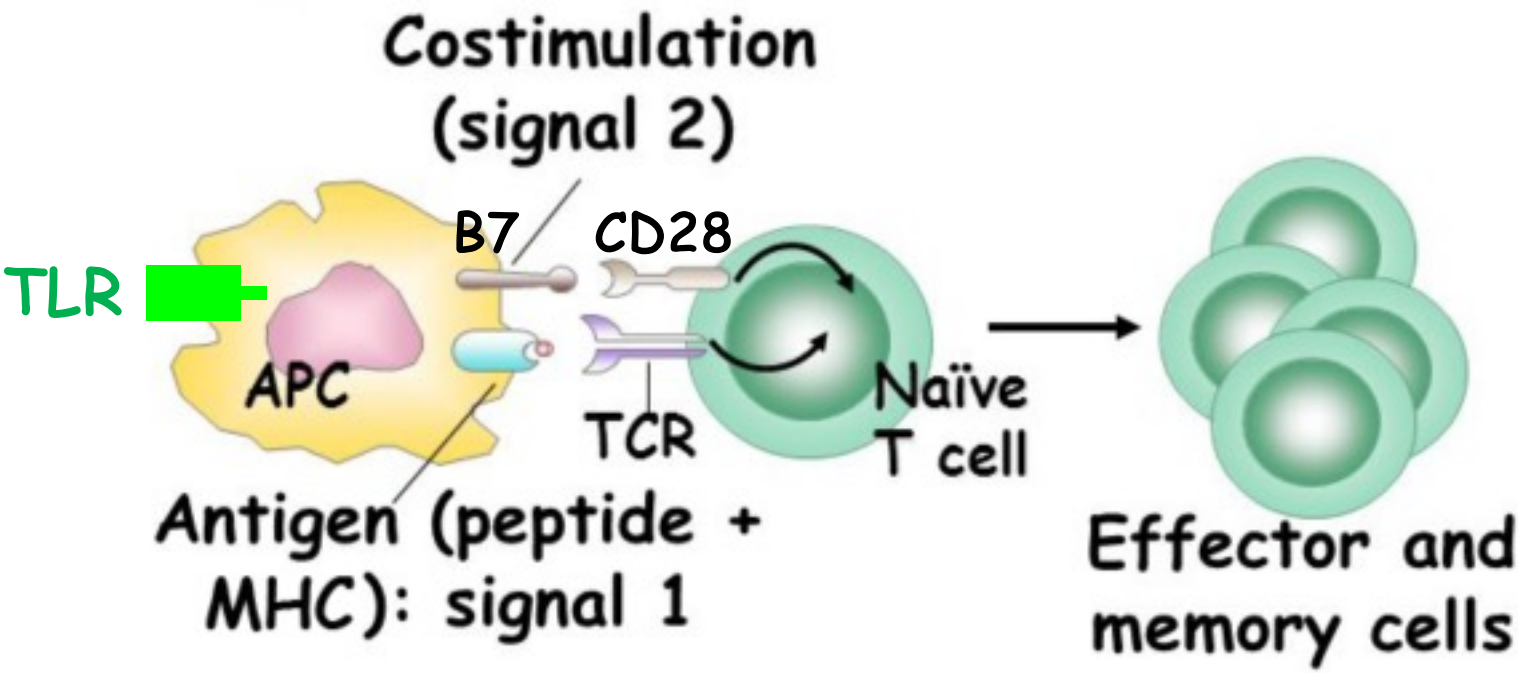


# Autoimmune Disease is caused by defective Immune Tolerance

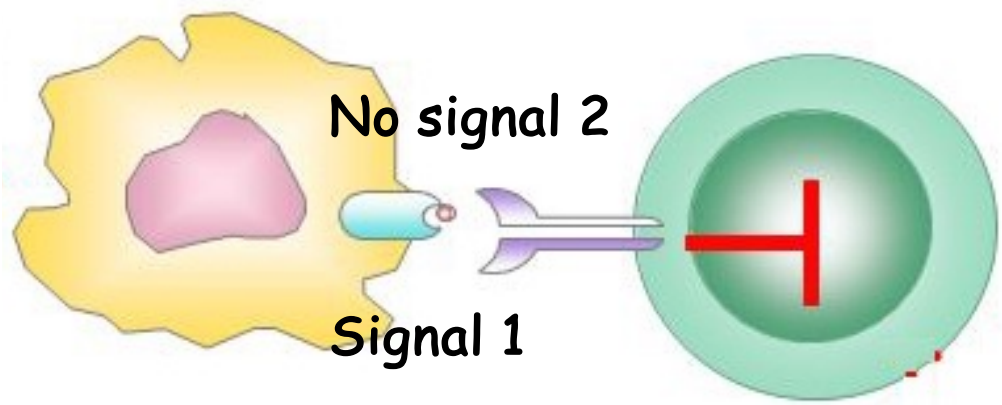


# Peripheral T cell Tolerance

***During immune response***

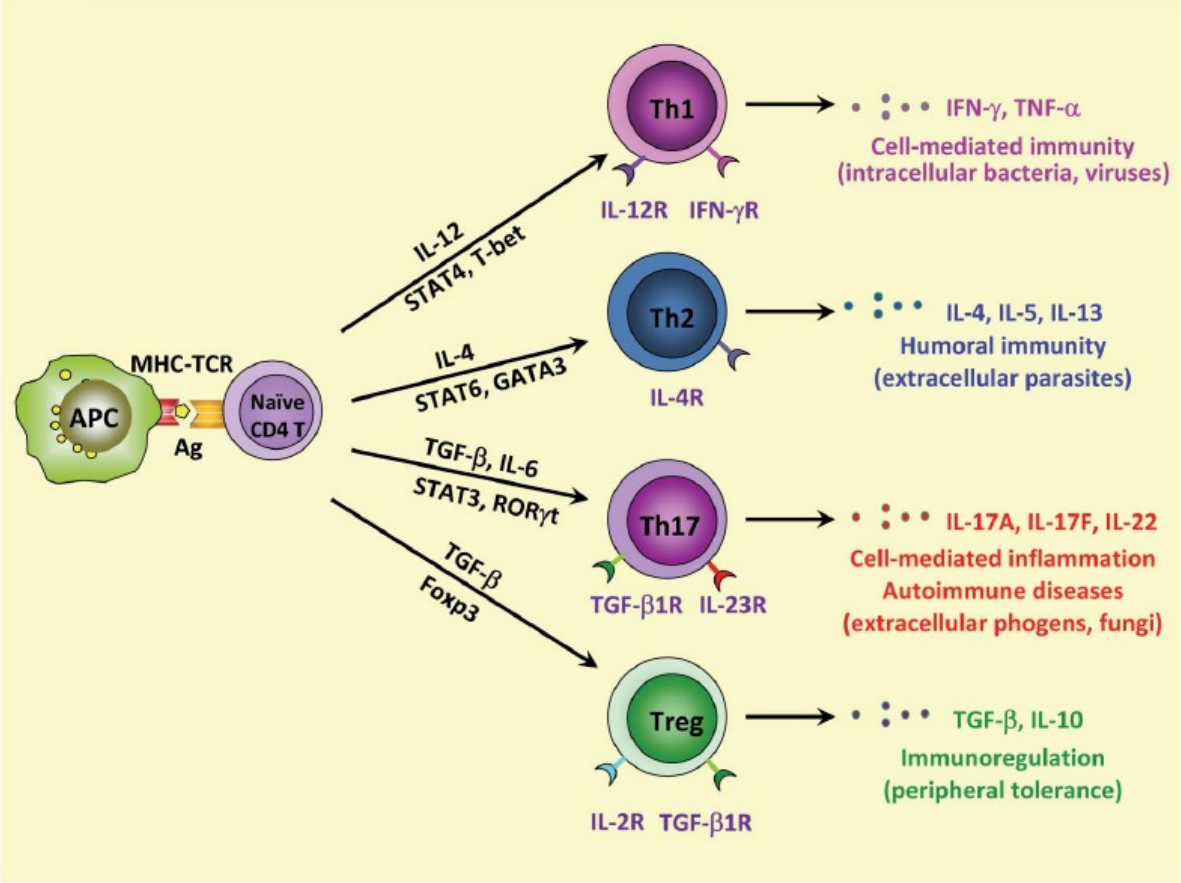


## *In the steady state*

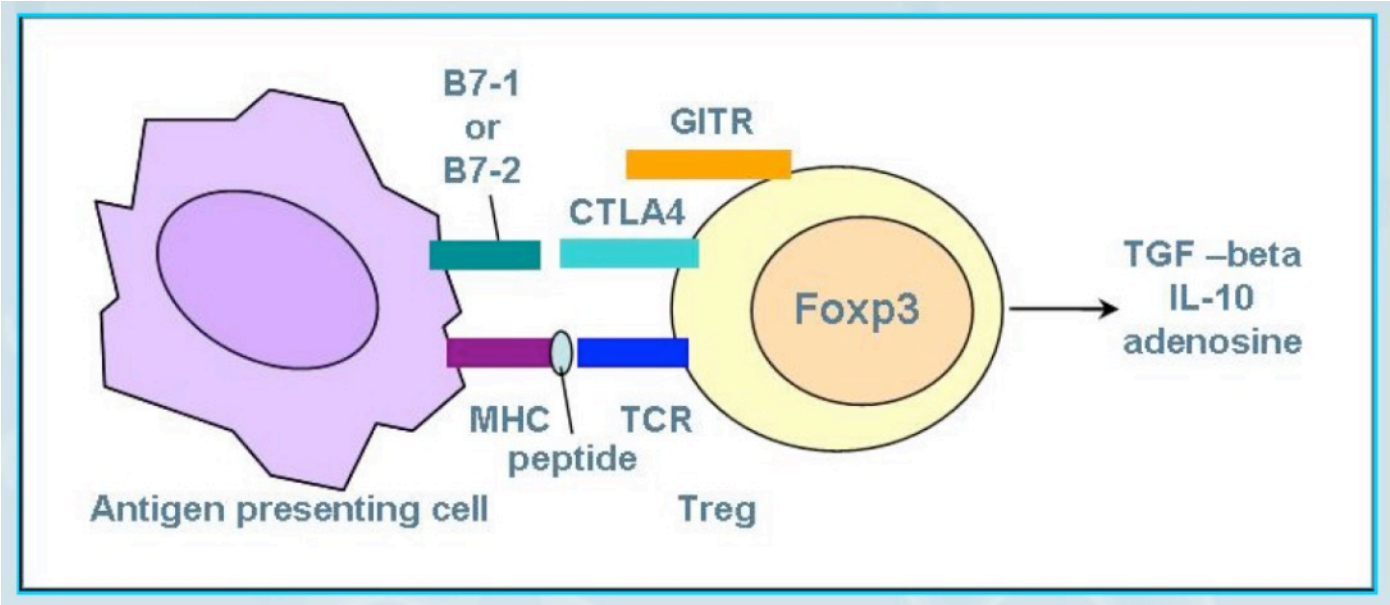



- Antigen recognition without adequate costimulation results in T cell anergy (functional inactivation) or death
  - **Peripheral Tolerance**

# CD4+ T cell Differentiation



# Tregs – mediators of peripheral tolerance




- Thought to interact with self-antigens through their TCR
- Express inhibitory receptors on their surface (CTLA4)
- Secrete immunosuppressive cytokines
- Mutations in Foxp3 cause IPEX syndrome

**Table 1: Characteristics of IPEX Syndrome #§.**

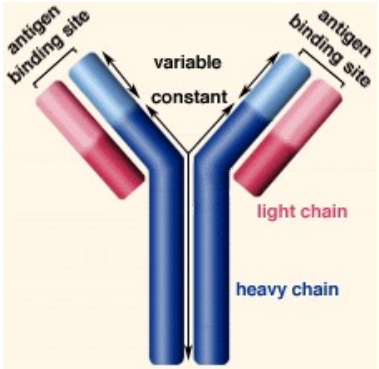
| Organ system | Manifestations* |
|---|---|
| Endocrine | Insulin dependent diabetes mellitus<br>Thyroid dysfunction<br>Parathyroid hormone resistance |
| Gastrointestinal | Autoimmune enteropathy<br>Diarrhea<br>Villous atrophy, failure to thrive |
| Skin | Eczema<br>Ichthyosiform dermatitis<br>Exfoliative dermatitis |
| Infectious disease | Exaggerated response<br>to viral infections<br>Frequent infections |
| Immune dysregulation | Increased IgE, intermittent eosinophilia<br>Skewing of T lymphocytes to Th2 phenotype<br>Hemolytic anemia<br>Immune thrombocytopenia<br>Coagulopathy |

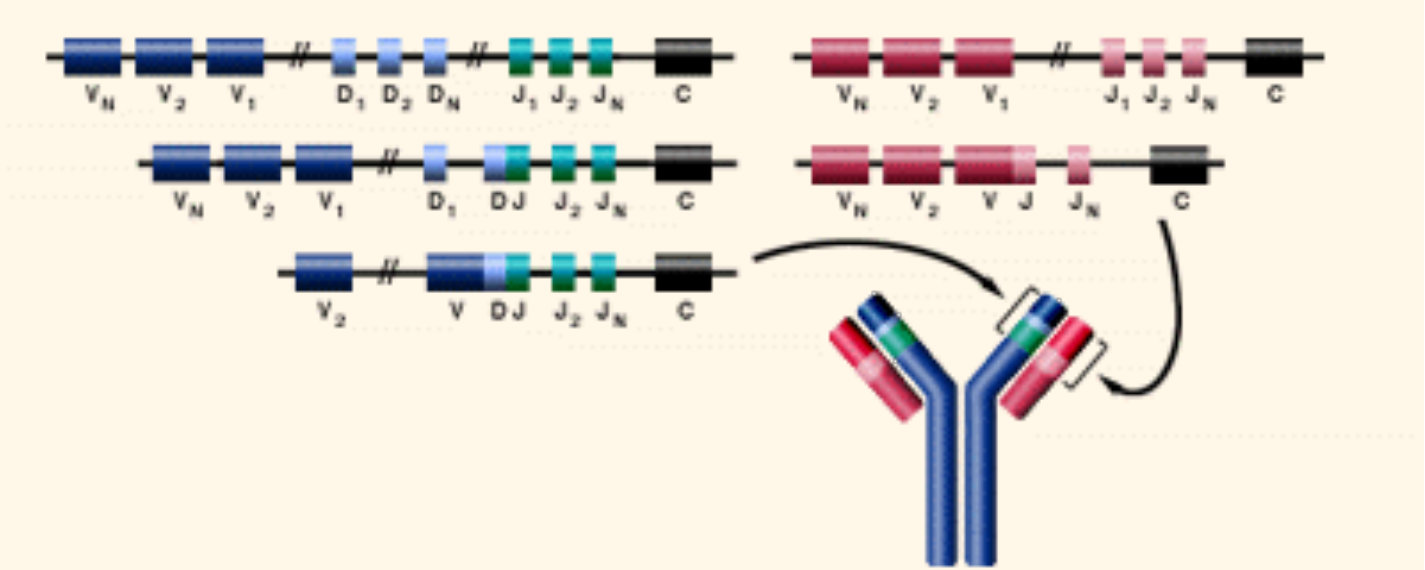
British Journal of Dermatology 2009 **160,** pp645–651

# B Cell Tolerance

In the bone marrow, pro- and pre-B cells rearrange their BCR genes (Immunoglobulin heavy and light chain genes) generating diversity.

**BCR**



# Central tolerance
# (during B cell development, in the bone marrow)

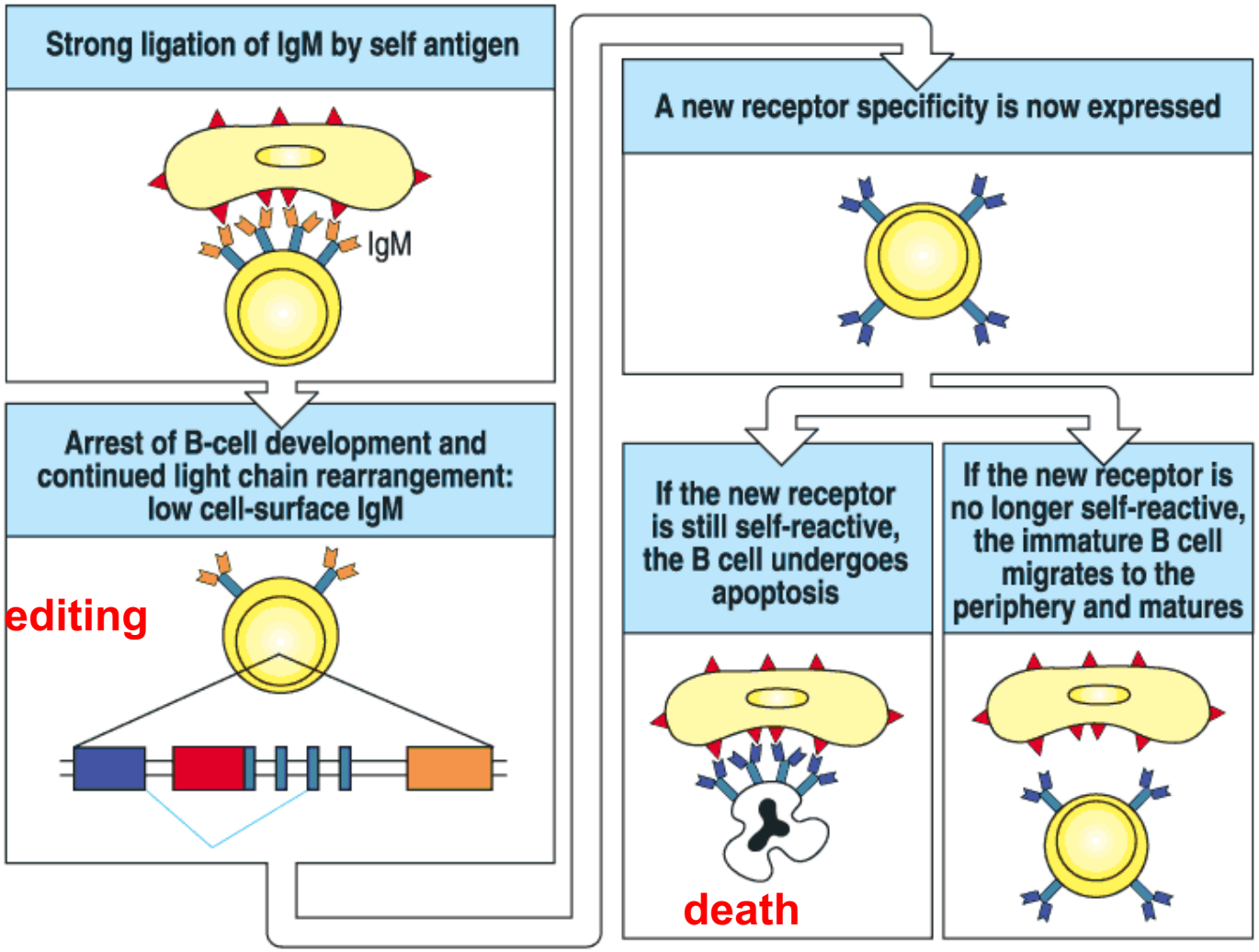



Fig 7.26 © 2001 Garland Science

# Peripheral tolerance
# (After B cell development, in the periphery)

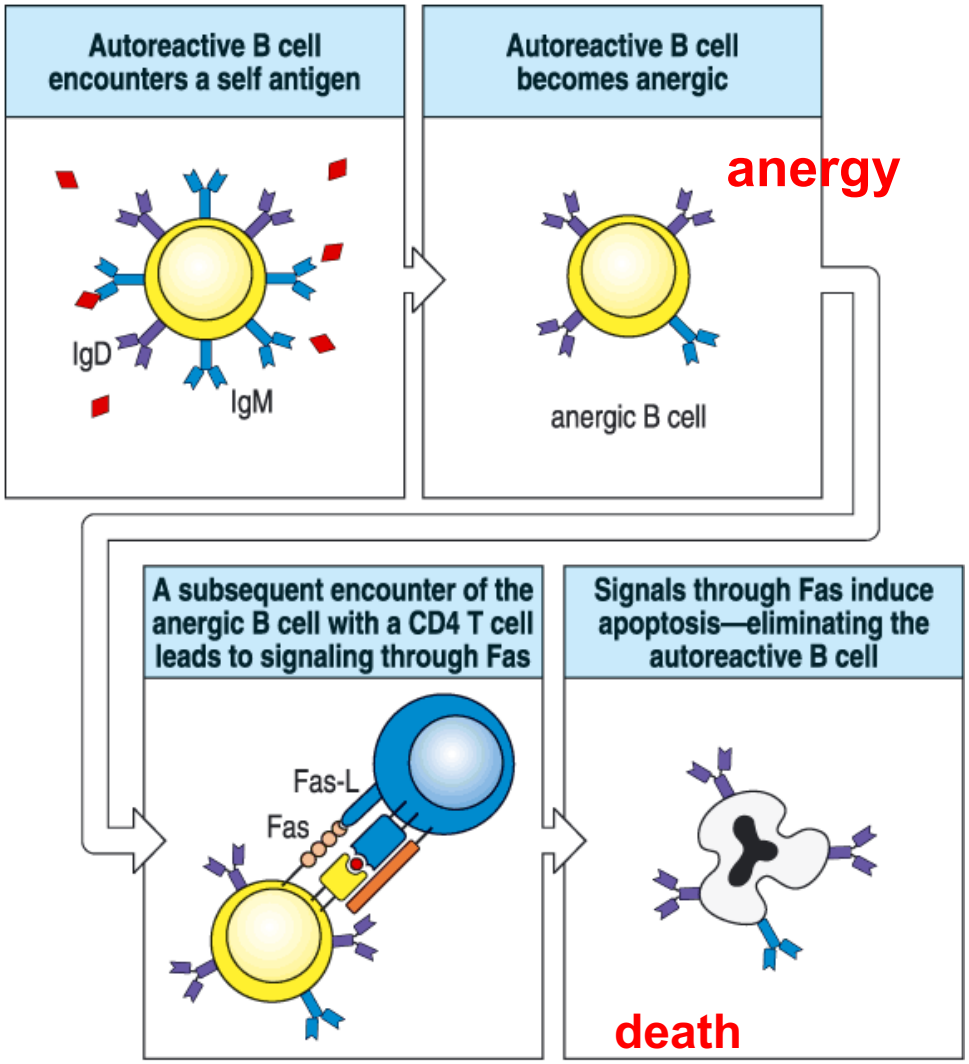



Fig 13.38 © 2001 Garland Science

# Autoimmunity

- A breakdown of tolerance
- There are inherited characteristics that confer susceptibility to autoimmunity

**HLA-associated risk factors for autoimmune disease**

| Disease | HLA allotype | Frequency (%) | | Relative risk |
|---|---|---|---|---|
| | | Patients | Control | |
| Ankylosing spondylitis | B27 | > 95 | 9 | > 150 |
| Birdshot chorioretinopathy | A29 | > 95 | 4 | > 50 |
| Narcolepsy | DQ6 | > 95 | 33 | > 40 |
| Celiac disease | DQ2 and DQ8 | 95 | 28 | 30 |
| Type 1 diabetes | DQ8 and DQ2 | 81 | 23 | 14 |
| Subacute thyroiditis | B35 | 70 | 14 | 14 |
| Multiple sclerosis | DQ6 | 86 | 33 | 12 |
| Rheumatoid arthritis | DR4 | 81 | 33 | 9 |
| Juvenile rheumatoid arthritis | DR8 | 38 | 7 | 8 |
| Psoriasis vulgaris | Cw6 | 87 | 33 | 7 |
| Addison's disease | DR3 | 69 | 27 | 5 |
| Graves' disease | DR3 | 65 | 27 | 4 |
| Myasthenia gravis | DR3 | 50 | 27 | 2 |
| Type 1 diabetes | DQ6 | < 0.1 | 33 | 0.02 |

Figure 13.24 The Immune System, 3ed. (© Garland Science 2009)

- Environmental/infectious agents have a significant role in triggering disease

- Damage can be focused (organ-specific) or systemic

- Types of Autoimmune Diseases

  1. T cell-dependent and organ-specific damage

  2. Antibody-dependent and organ-specific damage

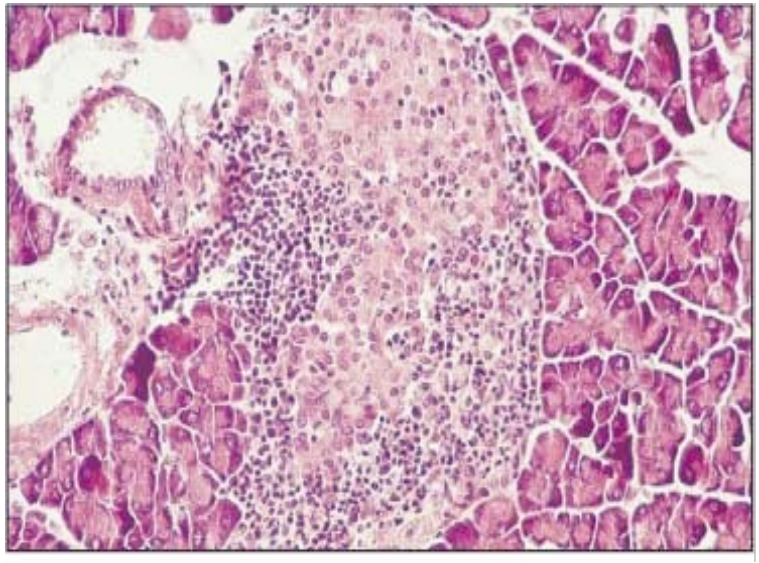
  3. Antibody-dependent and systemic damage

# 1. T Cell-Dependent and organ-specific

T cell-mediated damage is caused by cytokines, cytotoxicity, inflammation....

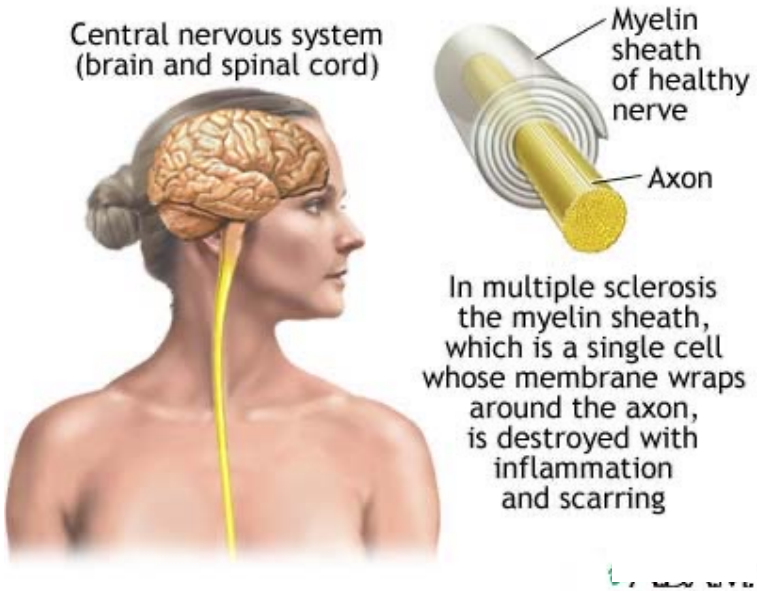
**eg. Insulin-dependent diabetes mellitus (type I diabetes)**

- Autoreactive T cells cause the destruction of the insulin-producing β cells in the Islets of Langerhans in the pancreas



# 1. T Cell-Dependent and organ-specific

## eg. Multiple Sclerosis




- Autoreactive T cells cause demyelination.

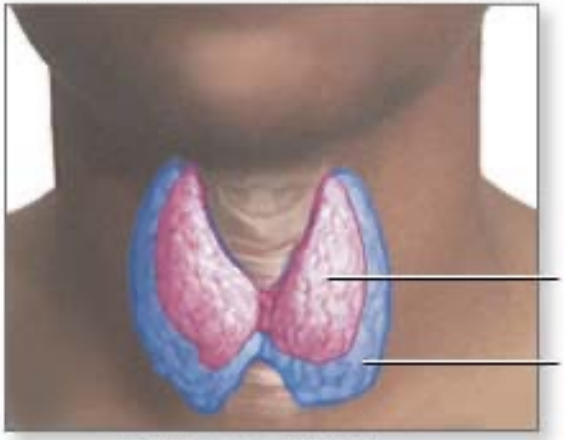
- A viral infection is thought to trigger the stimulation of autoreactive T cells.

# 2. Antibody-Dependent and organ-specific

## eg. Graves disease (hyperthyroidism)

- Antibodies to the receptor for thyroid stimulating hormone
- The antibodies stimulate the release of thyroid hormones

Exophthalmos (bulging eyes)




Diffuse goiter

Graves' disease is a common cause of hyperthyroidism, an over-production of thyroid hormone, which causes enlargement of the thyroid and other symptoms such as exophthalmos, heat intolerance and anxiety

ADAM.

# 2. Antibody-Dependent and organ-specific

## eg. Hashimoto's thyroiditis (hypothyroidism)

- Antibodies damage thyroid cells and interfere with their ability to make thyroid hormone.

- Over time, thyroid damage can cause thyroid hormone levels to be too low - hypothyroidism

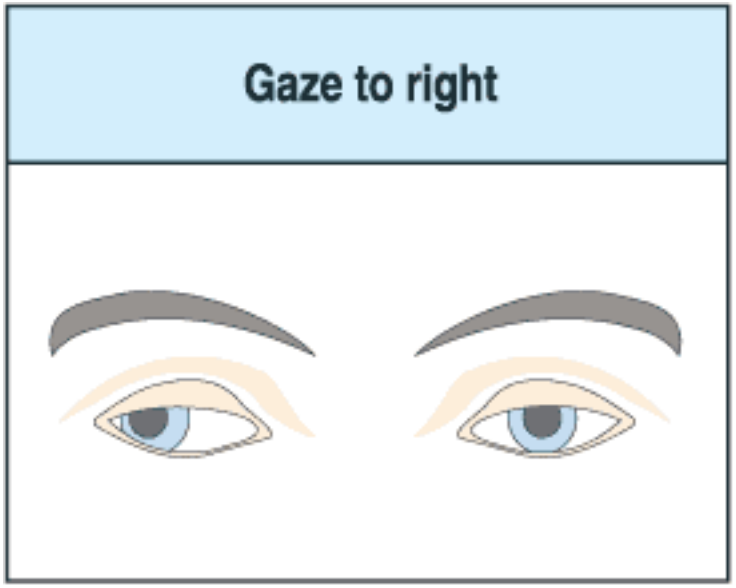
- An underactive thyroid causes every function of the body to slow down, such as heart rate, brain function, and the rate your body turns food into energy.

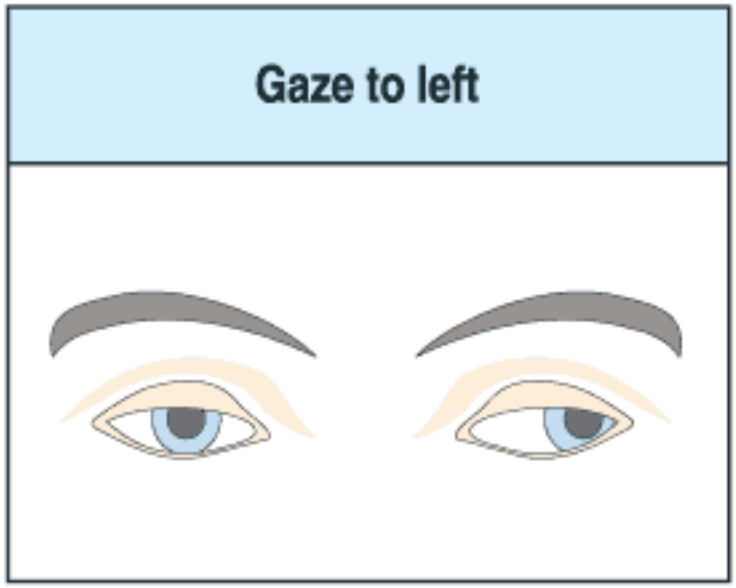
- An enlarged thyroid, called a goiter, is often the first sign of disease.

# 2. Antibody-Dependent and organ-specific

## eg. Myasthenia Gravis

- Ocular movement dysfunction – present in 50% of patients at presentation
- Muscle weakness

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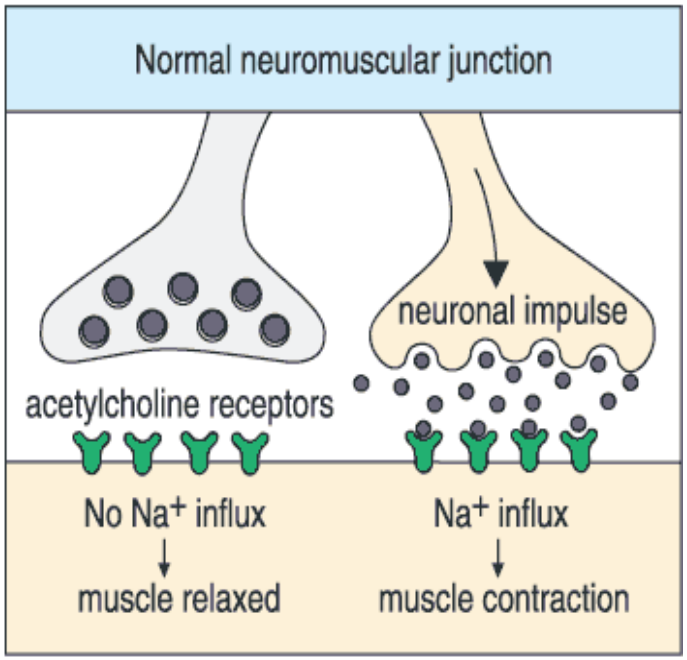
Muscle weakness by auto-antibodies against the acetylcholine receptor

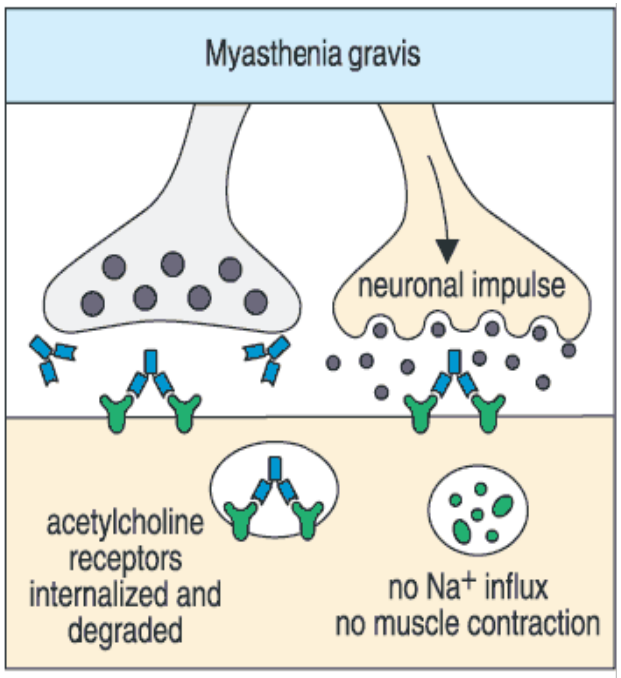
-block acetylcholine binding

-impair nerve impulse

-fix complement, recruit phagocytes

-damage

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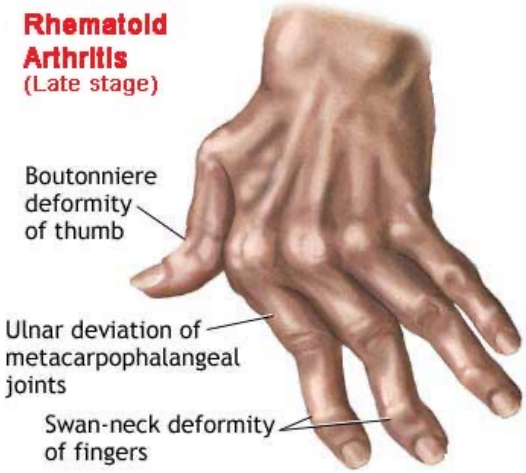
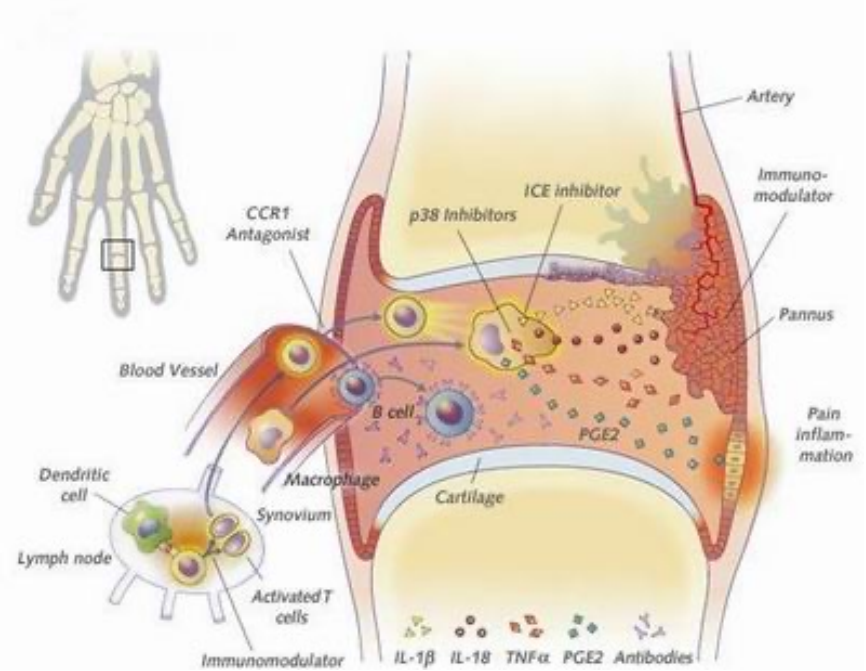
# 2. Antibody-Dependent and organ-specific

## eg. Rheumatoid Arthritis




- Autoantibodies are formed against Fc portion of IgG, called rheumatoid factor and anti-CCP autoantibodies

- People with HLA-DR4 genes are predisposed.

- Inflammation of the small joints of the hands and feet.

# 3. Antibody-dependent and systemic

eg. SLE ( Systemic lupus erythematosus)

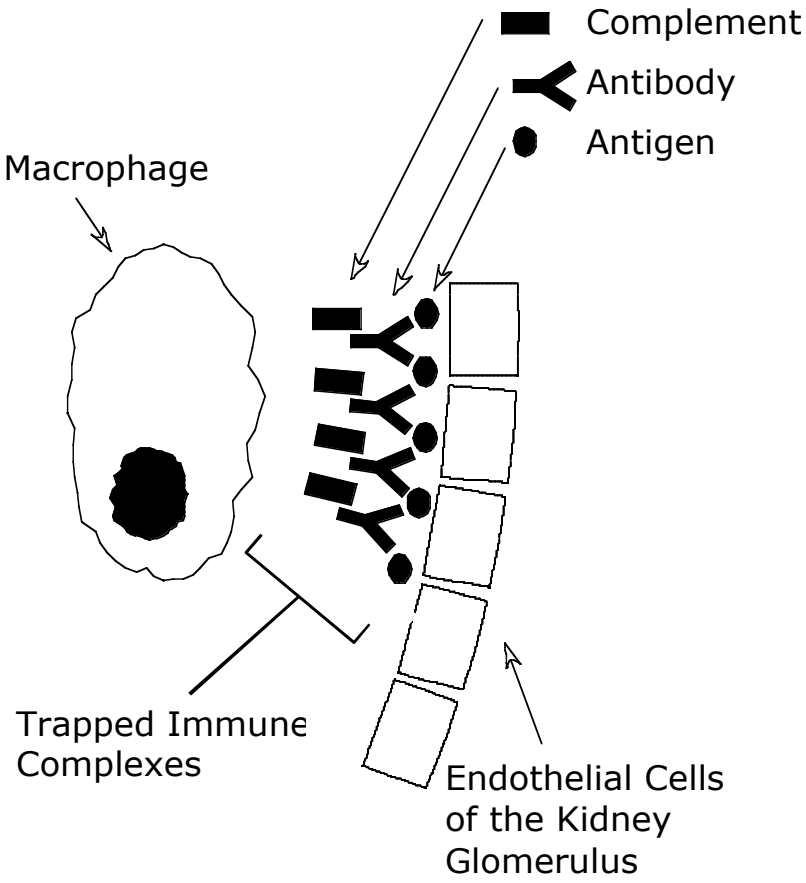
- Autoantibodies are formed against DNA, histones, nucleolar proteins, and other components of the cell nucleus.

- Antibodies against the double stranded DNA are the hallmark of SLE.

- Clinical features are caused by immune complexes that activate complement, as a consequence, damage tissues – can affect multiple organ systems – skin, joints, brain, lung, **kidney (glomerulonephritis)**

# glomerular nephritis :

Macrophages are attracted and bind to the immune complexes. They cause damage to the endothelium and basement membrane leading to kidney dysfunction



# Learning Objectives

- Describe what happens during "positive selection" of thymocytes and how this process links specificity of a T cell to its eventual effector function
- Describe what happens during "negative selection" of thymocytes and the role of the AIRE protein in this process.
- Describe central tolerance and specify why it is an imperfect/incomplete mechanism of preventing autoimmunity (i.e., what are its limitations).
- Describe the major mechanisms involved in peripheral tolerance specifically anergy, apoptosis, and suppression by regulatory T cells.
- Describe what happens during "negative selection" of B cells and the role of receptor editing during this process
- Describe the 3 major types of autoimmune diseases; for each category, provide an example of a disease, what the major clinical manifestations are and how the underlying immune defect contributes to the symptoms.